

Advanced Number Theory



Divisibility rules and prime factorization

- 1 Find the prime factorization of 1260.
 - 2 Determine whether 143 is prime, and justify your answer.
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GCF, LCM, and their relationship

- 3 Find the GCF and LCM of 36 and 60.
 - 4 Given that two numbers have GCF 8 and LCM 96, and one of the numbers is 24, find the other number, using the identity $\text{GCF} \times \text{LCM} = \text{product of the two numbers}$.
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Modular arithmetic

- 5 Find the remainder when 47 is divided by 6, and express this using modular notation.
 - 6 Find the last digit of 7^{100} , using the pattern of the last digits of powers of 7 (which cycle every 4: 7, 9, 3, 1).
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Sums and products of integer sets

- 7 Find the sum of all integers from 1 to 100, using the formula $S = \frac{n(n+1)}{2}$.
 - 8 Find the sum of all even integers from 2 to 200.
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Number sets and properties

- 9 Classify $\sqrt{16}$ as rational or irrational, and explain your reasoning.
 - 10 Explain why the sum of a rational number and an irrational number is always irrational.
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Sequences and series

- 11 Find the 15th term of the arithmetic sequence with first term 6 and common difference 4.

- 12 Find the sum of the first 8 terms of the geometric sequence 2, 6, 18, 54, ...
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Absolute value and inequalities with integers

- 13 Find all integer solutions of $|x - 5| \leq 3$.
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Combining number theory concepts in a multi-part problem

- 14 This question has three parts. (a) Find the prime factorization of 84 and 126. (b) Use the prime factorizations to find the GCF and LCM. (c) Verify your answer using the identity $\text{GCF} \times \text{LCM} = \text{product of the two numbers}$.
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Perfect squares, perfect cubes, and their properties

- 15 Find the smallest positive integer n such that $200n$ is a perfect square.
- 16 Determine whether 128 is a perfect cube, and justify your answer using prime factorization.
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Counting divisors of a number

- 17 Find the number of positive divisors of 360, using its prime factorization $360 = 2^3 \times 3^2 \times 5$.
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Working with consecutive integers and integer constraints

- 18 Find three consecutive even integers whose sum is 96.
- 19 The product of two consecutive positive integers is 156. Find the integers.
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Digit sum and place value problems

- 20 A two-digit number has digit sum 11. When the digits are reversed, the new number is 27 more than the original. Find the original number.
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A multi-part modular arithmetic and remainder problem

- 21 This question has three parts. (a) Find the remainder when 2^{10} is divided by 7. (b) Use the pattern of remainders of powers of 2 divided by 7 (which cycle) to find the remainder when 2^{50} is divided by 7. (c) Explain the general method for finding remainders of large powers using cycles.
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Multiple choice — easy

- 22 Find the remainder when 128 is divided by 9.
- a. 1
 - b. 9
 - c. 2
 - d. 3
- 23 Find the remainder when 88 is divided by 6.
- a. 6
 - b. 5
 - c. 3
 - d. 4
- 24 Find the remainder when 241 is divided by 12.
- a. 0
 - b. 12
 - c. 1
 - d. 2

25 Find the remainder when 180 is divided by 7.

- a. 4
- b. 6
- c. 7
- d. 5

26 Find the remainder when 78 is divided by 11.

- a. 11
- b. 0
- c. 2
- d. 1

27 Find the remainder when 80 is divided by 7.

- a. 2
- b. 7
- c. 3
- d. 4

28 Find the remainder when 205 is divided by 9.

- a. 8
- b. 9
- c. 7
- d. 6

29 Find the remainder when 183 is divided by 9.

- a. 2
- b. 3
- c. 9
- d. 4

30 Find the remainder when 35 is divided by 11.

- a. 2
- b. 3
- c. 1
- d. 11

31 Find the remainder when 71 is divided by 3.

- a. 3
- b. 2
- c. 0
- d. 1

32 Find the remainder when 78 is divided by 10.

- a. 7
- b. 10
- c. 9
- d. 8

33 Find the remainder when 34 is divided by 10.

- a. 5
- b. 3
- c. 4
- d. 10

34 Find the remainder when 74 is divided by 13.

- a. 8
- b. 9
- c. 10
- d. 13

35 Find the remainder when 226 is divided by 13.

- a. 4
- b. 6
- c. 5
- d. 13

36 Find the remainder when 177 is divided by 8.

- a. 2
- b. 1
- c. 8
- d. 0

Multiple choice — medium

37 Find the LCM of 24 and 36.

- a. 72
- b. 864
- c. 84
- d. 12

38 Find the LCM of 54 and 12.

- a. 648
- b. 6
- c. 108
- d. 114

39 Find the LCM of 40 and 50.

- a. 200
- b. 2000
- c. 10
- d. 210

40 Find the LCM of 12 and 18.

- a. 6
- b. 36
- c. 216
- d. 42

41 Find the LCM of 20 and 50.

- a. 110
- b. 10
- c. 1000
- d. 100

42 Find the LCM of 24 and 18.

- a. 432
- b. 72
- c. 78
- d. 6

43 Find the LCM of 189 and 147.

- a. 21
- b. 27783
- c. 1323
- d. 1344

44 Find the LCM of 375 and 135.

- a. 50625
- b. 3390
- c. 15
- d. 3375

45 Find the LCM of 75 and 45.

- a. 15
- b. 225
- c. 240
- d. 3375

46 Find the LCM of 100 and 250.

- a. 500
- b. 25000
- c. 50
- d. 550

47 Find the LCM of 100 and 40.

- a. 4000
- b. 20
- c. 200
- d. 220

48 Find the LCM of 250 and 200.

- a. 1050
- b. 50000
- c. 1000
- d. 50

49 Find the LCM of 20 and 250.

- a. 500
- b. 10
- c. 5000
- d. 510

50 Find the LCM of 225 and 135.

- a. 675
- b. 30375
- c. 45
- d. 720

51 Find the LCM of 175 and 245.

- a. 1225
- b. 42875
- c. 1260
- d. 35

Multiple choice — hard

52 Find the 14th term of the arithmetic sequence with first term -4 and common difference 3.

- a. 38
- b. 32
- c. 37
- d. 35

- 53 Find the 19th term of the arithmetic sequence with first term 7 and common difference 7.
- a. 133
 - b. 140
 - c. 130
 - d. 136
- 54 Find the 7th term of the geometric sequence 2, 4, 8, 16, ...
- a. 130
 - b. 125
 - c. 131
 - d. 128
- 55 Find the 4th term of the geometric sequence 3, 6, 12, 24, ...
- a. 27
 - b. 24
 - c. 21
 - d. 26
- 56 Find the 5th term of the geometric sequence 1, 2, 4, 8, ...
- a. 19
 - b. 13
 - c. 18
 - d. 16

- 57 Find the number of positive divisors of 2450, using its prime factorization $2450 = 2 \times 7^2 \times 5^2$.
- a. 16
 - b. 20
 - c. 18
 - d. 4
- 58 Find the number of positive divisors of 528, using its prime factorization $528 = 2^4 \times 3 \times 11$.
- a. 18
 - b. 20
 - c. 22
 - d. 4
- 59 Find the number of positive divisors of 315, using its prime factorization $315 = 3^2 \times 7 \times 5$.
- a. 12
 - b. 14
 - c. 10
 - d. 2
- 60 Find the number of positive divisors of 900, using its prime factorization $900 = 2^2 \times 3^2 \times 5^2$.
- a. 25
 - b. 8
 - c. 29
 - d. 27

- 61 Find the number of positive divisors of 140, using its prime factorization $140 = 2^2 \times 7 \times 5$.
- a. 12
 - b. 2
 - c. 14
 - d. 10
- 62 Find the last digit of 3^{24} , using the pattern of the last digits of powers of 3 (which cycle every 4: 3, 9, 7, 1).
- a. 7
 - b. 3
 - c. 1
 - d. 9
- 63 Find the last digit of 3^{100} , using the pattern of the last digits of powers of 3 (which cycle every 4: 3, 9, 7, 1).
- a. 7
 - b. 3
 - c. 1
 - d. 9
- 64 Find the last digit of 3^{132} , using the pattern of the last digits of powers of 3 (which cycle every 4: 3, 9, 7, 1).
- a. 7
 - b. 9
 - c. 1
 - d. 3

- 65 Find the last digit of 3^{188} , using the pattern of the last digits of powers of 3 (which cycle every 4: 3, 9, 7, 1).
- a. 1
 - b. 7
 - c. 9
 - d. 3
- 66 Find the last digit of 4^{129} , using the pattern of the last digits of powers of 4 (which cycle every 2: 4, 6).
- a. 4
 - b. 6
 - c. 7
 - d. 9